

A full sweep–Haar-multiplier characterization of semicommutative martingale paraproducts

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19 July 2026

Status. Literature-implied full structural answer, likely valid, subject to human review. The dyadic operator-valued theorem is due to Blasco and Pott [2, Lemma 2.3 and Theorem 2.4, pp. 6–7]. We identify its semicommutative consequence and give the direct d -adic proof.

1 Source question and answer

Wei and Zhang ask how to characterize boundedness of a semicommutative martingale paraproduct on L_2 [1, p. 2]:

“It still remains open how to characterize the boundedness of π_b .”

The surrounding passage is reproduced below.

Nazarov, Treil and Volberg also independently obtained (1.1) in [16] by the Bellman method, and they also gave an example showing that for any $n \in \mathbb{N}$ there exists b such that

$$\|\pi_b\|_{B(L_2(L_\infty(\mathbb{R}) \otimes \mathbb{M}_n))} \gtrsim \sqrt{\log(n+1)} \|b\|_{BMO_{so}^2(\mathbb{R}, \mathbb{M}_n)}.$$

This implies that the boundedness of π_b cannot be characterized solely by $BMO_{so}^2(\mathbb{R}, \mathcal{M})$ for infinite-dimensional $\mathcal{M}$. In [15], Nazarov, Pisier, Treil and Volberg proved that $\log(n+1)$ is the optimal order of the best constant in (1.1). It still remains open how to characterize the boundedness of π_b on $L_2(L_\infty(\mathbb{R}) \otimes \mathcal{M})$.

Although we do not know how to describe the boundedness of semicommutative martingale paraproducts until now, some particular properties of π_b can be investigated. For instance, Mei succeeded in describing the extrapolation property by virtue of $BMO_{\mathcal{M}}^2(\mathbb{R})$ for dyadic martingale paraproducts in [13]. Recall that $BMO_{\mathcal{M}}^d(\mathbb{R})$ is the operator-valued BMO space associated with the d -adic martingales consisting of all $\mathcal{M}$ -valued functions b that are Bochner integrable on any d -adic interval such that

$$(1.2) \quad \|b\|_{BMO_{\mathcal{M}}^d(\mathbb{R})} = \sup_{I \in \mathcal{D}} \left(\frac{1}{m(I)} \int_I \|b - (\frac{1}{m(I)} \int_I b \, dm)\|_{\mathcal{M}}^2 \, dm \right)^{1/2} < \infty.$$

There is a full answer if one allows the natural positive sweep of the symbol and the independently defined Haar-multiplier BMO norm. For arbitrary noncommuting symbols,

$$\pi_b \text{ is bounded} \iff b \in BMO_c \text{ and } S_b \in BMO_{\text{mult}}.$$

The extra sweep condition is essential: the known logarithmic matrix examples rule out a characterization by column BMO alone.

2 Definitions

Fix $d \geq 2$ and a d -adic grid $\mathcal{D}$ on $\mathbb{R}$. For every $I \in \mathcal{D}$, choose a real orthonormal basis $h_I^1, \dots, h_I^{d-1}$ of the functions supported on I , constant on its d children, and of integral zero. Let $(\mathcal{M}, \tau)$ be a semifinite von Neumann algebra and put

$$\mathcal{N} = L_\infty(\mathbb{R}) \overline{\otimes} \mathcal{M}, \quad H = L_2(\mathcal{M}, \tau), \quad L_2(\mathcal{N}) = L_2(\mathbb{R}; H).$$

For a locally integrable $\mathcal{M}$ -valued symbol b , write

$$b_I^r = \int_I b(t) h_I^r(t) dt, \quad a_I = \sum_{r=1}^{d-1} (b_I^r)^* b_I^r \in \mathcal{M}_+.$$

If a coefficient is not bounded, all norms below are interpreted as $+\infty$. This causes no loss: boundedness of the paraproduct forces every coefficient to define a bounded left multiplier on H , hence to belong to $\mathcal{M}$.

For a finite set $F \subseteq \mathcal{D}$, define

$$\begin{aligned} \pi_b^F f &= \sum_{I \in F} \sum_{r=1}^{d-1} h_I^r b_I^r m_I f, & m_I f &= \frac{1}{|I|} \int_I f(t) dt, \\ S_F b(t) &= \sum_{I \in F} a_I \frac{\mathbf{1}_I(t)}{|I|}. \end{aligned}$$

For any self-adjoint formal symbol c , put

$$\Lambda_c = \pi_c + \pi_c^*, \quad \pi_c f = \sum_{J \in \mathcal{D}} \sum_{s=1}^{d-1} h_J^s c_J^s m_J f, \quad c_J^s = \int_J c(t) h_J^s(t) dt.$$

For finite F , $S_F b$ is a bounded finite-step symbol, so $\Lambda_{S_F b}$ is unambiguously defined. Set

$$P(b) = \sup_{F \subseteq \mathcal{D}} \|\pi_b^F\|_{2 \rightarrow 2}, \tag{1}$$

$$M(b) = \sup_{F \subseteq \mathcal{D}} \|\Lambda_{S_F b}\|_{2 \rightarrow 2}, \tag{2}$$

$$C(b)^2 = \sup_{K \in \mathcal{D}} \left\| \frac{1}{|K|} \sum_{I \subseteq K} a_I \right\|_{\mathcal{M}}. \tag{3}$$

Positive sums in (3) are understood as suprema of their finite partial sums. Up to the harmless one-level indexing convention, $C(b) = \|b\|_{\text{BMO}_{c,d}}$ in the source paper. The quantity $M(b)$ is the uniform d -adic Haar-multiplier norm of the formal sweep S_b .

3 Idea of the proof

The product $\pi_b^* \pi_b$ is much more rigid than π_b . Orthogonality of the Haar differences turns it into the positive dyadic averaging operator

$$f \mapsto \sum_I \frac{\mathbf{1}_I}{|I|} a_I m_I f.$$

Relative to the Haar-tree decomposition, its strict lower triangular part is the paraproduct with symbol S_b and its strict upper triangular part is the adjoint. The remaining block diagonal is positive. On a Haar block over J , every vector is constant on the children of J ; grouping descendants child by child bounds the diagonal block exactly by $C(b)^2$. Indicator tests give the reverse necessary bound $C(b) \leq P(b)$. No commutation of the coefficients is used.

4 Full characterization

Theorem 1 (Sweep-multiplier characterization). *With the definitions above,*

$$P(b)^2 \leq M(b) + C(b)^2 \leq 3P(b)^2. \quad (4)$$

Consequently the formal semicommutative paraproduct extends boundedly to $L_2(\mathcal{N})$ if and only if

$$b \in \text{BMO}_{c,d} \quad \text{and} \quad \sup_{F \in \mathcal{D}} \|\Lambda_{S_F b}\|_{2 \rightarrow 2} < \infty.$$

When these conditions hold, the norm of the extension is $P(b)$. The constants in (4) are independent of d , of $\mathcal{M}$, and of matrix dimension.

Proof. Fix $F \in \mathcal{D}$ and abbreviate $\pi_F = \pi_b^F$. Haar orthogonality gives

$$\pi_F^* \pi_F f = \sum_{I \in F} \frac{\mathbf{1}_I}{|I|} a_I m_I f. \quad (5)$$

Indeed, the I, r Haar coefficient of $\pi_F f$ is $b_I^r m_I f$, and the adjoint of $m_I : H\text{-valued } L_2 \rightarrow H$ sends x to $\mathbf{1}_I x / |I|$.

Let $\mathcal{H}_J$ be the $(d-1)$ -dimensional scalar Haar space over J . Then

$$L_2(\mathbb{R}; H) = \bigoplus_{J \in \mathcal{D}} (\mathcal{H}_J \otimes H).$$

Taking matrix blocks of (5), disjoint Haar intervals give zero. If $K \subsetneq J$, a direct coefficient computation gives

$$P_K(\pi_F^* \pi_F) P_J = P_K \pi_{S_F b} P_J,$$

and the reverse nesting gives the corresponding block of $\pi_{S_F b}^*$. Thus

$$\pi_F^* \pi_F = \Lambda_{S_F b} + D_F, \quad (6)$$

where D_F is the block diagonal of the positive operator $\pi_F^* \pi_F$ relative to the displayed Haar decomposition. This is the d -adic version of the Blasco–Pott product identity [2, Lemma 2.3].

We next bound the diagonal. An element $u \in \mathcal{H}_J \otimes H$ is constant, say equal to $x_K \in H$, on every child K of J . By (5),

$$\begin{aligned} \langle D_F u, u \rangle &= \sum_{K \text{ child of } J} \sum_{\substack{I \in F \\ I \subseteq K}} \langle a_I x_K, x_K \rangle_H \\ &\leq C(b)^2 \sum_{K \text{ child of } J} |K| \|x_K\|_H^2 = C(b)^2 \|u\|_2^2. \end{aligned}$$

Here a_I acts by left multiplication, and (3) says $\sum_{I \subseteq K} a_I \leq |K|C(b)^2 \mathbf{1}$ as positive operators. Therefore

$$0 \leq D_F \leq C(b)^2 \mathbf{1}. \quad (7)$$

Equation (6) and (7) yield

$$\|\pi_F\|^2 \leq \|\Lambda_{S_F b}\| + C(b)^2. \quad (8)$$

Taking the supremum over F proves the first inequality in (4).

For the converse estimates, fix $K \in \mathcal{D}$ and $x \in H$. Apply π_F to $|K|^{-1/2} \mathbf{1}_K x$ and retain only output Haar coefficients indexed by $I \subseteq K$. Orthogonality gives

$$\left\langle \frac{1}{|K|} \sum_{\substack{I \in F \\ I \subseteq K}} a_I x, x \right\rangle_H \leq \|\pi_F\|^2 \|x\|_H^2.$$

Taking suprema over x, K, F gives

$$C(b)^2 \leq P(b)^2. \quad (9)$$

Moreover, (6)–(7) imply

$$\|\Lambda_{S_F b}\| \leq \|\pi_F^* \pi_F\| + \|D_F\| \leq 2P(b)^2.$$

Hence $M(b) + C(b)^2 \leq 3P(b)^2$, proving (4).

If the full paraproduct is bounded, every π_F is an orthogonal output projection of it, so $P(b)$ is at most its norm. Conversely, if $P(b) < \infty$, the orthogonal series defining $\pi_b f$ converges for every finite Haar polynomial f and has norm at most $P(b)\|f\|_2$. Density gives a bounded extension of norm $P(b)$. This finishes the proof. $\square$

Corollary 2 (Explicit positive lower-tree multiplier). *For finite F , define*

$$\Phi_{J,F}(t) = \sum_{\substack{I \in F \\ I \subsetneq J}} a_I \frac{\mathbf{1}_I(t)}{|I|}$$

and, on finite Haar expansions,

$$Q_{b,F} \left(\sum_{J,s} h_J^s x_J^s \right) = \sum_{J,s} h_J^s \Phi_{J,F} x_J^s.$$

Then $Q_{b,F} = \pi_F^ \pi_F$ and*

$$\|\pi_F\|^2 = \|Q_{b,F}\|.$$

Thus the criterion can equivalently be checked through an explicit positive lower-tree Haar multiplier built only from the square coefficients a_I .

Proof. Apply (5) to $h_J^s x$. Its average over I is zero unless $I \subsetneq J$, in which case $m_I(h_J^s x) = h_J^s(t)x$ on I . This is precisely the displayed formula. $\square$

5 Why the extra condition is genuine

Testing π_b only on one filtration level recovers (3), i.e. column BMO. The finite-dimensional examples of Katz and of Nazarov–Pisier–Treil–Volberg show that the best general estimate has logarithmic matrix growth [3]. If

$$\|\pi_b\| \simeq \log(n+1) C(b),$$

then (4) forces

$$M(b) \gtrsim (\log(n+1))^2 C(b)^2.$$

The sweep multiplier therefore records exactly the multiscale noncommutative interaction invisible to projection or single-level BMO tests. It is not column BMO under another name.

If the range of b lies in an abelian von Neumann subalgebra, simultaneous spectral decomposition reduces all fibers to the scalar Carleson embedding. Then $M(b) \lesssim_d C(b)^2$, and Theorem 1 recovers the earlier commuting-symbol packet as a corollary.

6 Provenance and exact scope

Blasco and Pott define the operator-valued paraproduct space BMO_{para} and Haar-multiplier space BMO_{mult} . Their Theorem 2.4 states, for dyadic $B(\mathcal{H})$ -valued symbols,

$$\|S_B\|_{\text{BMO}_{\text{mult}}} + \|B\|_{\text{SBMO}^d}^2 \asymp \|\pi_B\|^2.$$

Taking $\mathcal{H} = L_2(\mathcal{M})$ and $B(t) = L_{b(t)}$ turns their vector paraproduct into the semicommutative left paraproduct and turns $B_I^* B_I$ into $L_{b_I^* b_I}$. This already gives the $d = 2$ answer. The proof above makes the d -adic extension explicit and handles all $(d-1)$ Haar modes in the diagonal-block estimate.

Accordingly, this packet does not claim a new dyadic theorem. It records a full literature-implied structural answer that appears not to have been connected to the 2024 source question. If “characterization” is intended to mean a criterion involving only a familiar scalar-like BMO oscillation norm, that narrower problem remains open; Theorem 1 shows precisely what extra positive multiplier datum is required.

References

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- [2] O. Blasco and S. Pott, *Operator-valued dyadic BMO spaces*, arXiv:0805.0158 (2008), published in J. Operator Theory 63 (2010), 333–347.
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